HACKATHON TRACK A — FULL TECHNICAL SPECIFICATION

CardioVision AI

Interactive Cardiovascular Risk Visualization & Prediction System with 360° Anatomical Modeling

ARCHITECTURE	ML CLASSIFIER	DATASET SIZE	VALIDATION SCORE
FastAPI + React 19 + Three.js	Calibrated HistGradientBoosting	70,000 Records (68,637 Cleaned)	ROC-AUC: 0.8053 | Acc: 73.29%

1. Executive Summary & Core Engineering Mission

CardioVision AI is a dual-tier medical software prototype that bridges predictive supervised machine learning with clinical 3D anatomical visualization. In conventional healthcare informatics, cardiovascular risk engines (e.g., ASCVD, Framingham) present outputs as abstract numerical probabilities that fail to engage patients or illustrate physical hemodynamics. CardioVision AI solves this communication bottleneck by predicting cardiovascular risk from multi-factorial clinical vitals and projecting those statistical findings directly onto an interactive, 360° rotating anatomical human heart model formatted to the gold-standard *Encyclopaedia Britannica* medical reference layout.

Foundational First Principle: Absolute Scientific Rigor

Following strict clinical integrity guidelines: **The underlying dataset target is binary cardiovascular disease presence/absence (`cardio`), NOT coronary artery disease (CAD), coronary stenosis, arterial narrowing, or plaque occlusion.** CardioVision AI strictly identifies its output as *"Cardiovascular Disease Risk Prediction"* and explicitly documents its 3D cardiovascular model as an *AI-generated educational visualization* rather than a patient-specific imaging scan (CT/MRI/Angiogram).

2. Complete Technology Stack & Architectural Taxonomy

The platform is engineered as a decoupled client-server architecture designed for high inference throughput, strict schema validation, and 60 FPS WebGL rendering on macOS / Apple Silicon:

TIER	TECHNOLOGY	VERSION	ENGINEERING ROLE & PURPOSE
Frontend Core	React	19.2.8	Declarative reactive UI, state management, and component hierarchy.
Build Tooling	Vite	8.3.3	Hot Module Replacement (HMR), tree-shaking, and ES module bundling.
3D Graphics	Three.js & React Three Fiber	three 0.186 / R3F 9.8	Hardware-accelerated WebGL rendering, 3D scenes, lighting, and camera.
3D Ecosystem	@react-three/drei	10.7.9	GLTF loading (`useGLTF`), OrbitControls, and canvas helpers.
Charting	Recharts	3.10.1	Feature importance horizontal bar charts and radar comparison plots.
Iconography	Lucide React	1.52.0	Clean, standardized medical and analytical vector icons.
Styling	Pure Vanilla CSS3	CSS Custom Props	Custom design system: serene light pastel clinical theme, zero Tailwind.
Backend API	FastAPI	0.115+	High-throughput, asynchronous RESTful API framework with Swagger OpenAPI.
ASGI Server	Uvicorn	0.32+	High-performance asynchronous server running on `http://0.0.0.0:8000`.
Data Validation	Pydantic v2	2.10+	Type coercion, numeric bounds, and cross-field systolic/diastolic validator.
Machine Learning	scikit-learn	1.6.0	Preprocessing pipelines, ColumnTransformer, ensemble modeling, and calibration.
Data Wrangling	pandas & numpy	2.2+ / 2.1+	Dataset cleaning, vectorized feature engineering, matrix math.
Serialization	joblib	1.4+	Serializing trained pipelines (`final_model.joblib`, 1.1 MB).

3. Dataset Audit, Preprocessing & Feature Engineering Pipeline

3.1 Dataset Verification & Clinical Bounds

The model is trained on the verified `cardio_train (1).csv` dataset comprising **70,000 raw patient records** and 13 attributes. Before model training, an exhaustive audit evaluated column types, missingness, and physiological plausibility:

- **Zero Missing Values:** The dataset contained no NaN or null values across all 70,000 records.
- **Class Balance:** Exactly 35,021 negative cases (50.03%) and 34,979 positive cases (49.97%), ensuring zero class imbalance bias.
- **Physiological Outlier Removal:** Data entry anomalies were eliminated using medical plausibility rules ($60 \le ap_hi \le 250$, $30 \le ap_lo \le 200$, $ap_lo < ap_hi$, $100 \le height \le 220$, $30 \le weight \le 250$, $10 \le bmi \le 80$). Exactly **1,363 rows (1.95%)** were pruned, retaining **68,637 valid clinical rows**.

3.2 Feature Conversions & Physiological Feature Engineering

1. Chronological Age

Converted from raw days to solar years: $ ext{age} = ext{age_days} / 365.25$. Range: 29.6 to 64.9 years.

2. Body Mass Index (BMI)

Computed as $ ext{weight} / (ext{height}/100)^2$. Normalizes mass against stature into WHO obesity classes.

3. Pulse Pressure & MAP

$PP = ap_hi - ap_lo$
$MAP = ap_lo + rac{1}{3}(ap_hi - ap_lo)$
Quantifies arterial stiffness and organ perfusion pressure.

3.3 Leakage-Proof Scikit-Learn Preprocessing Architecture

To ensure mathematical reproducibility and eliminate data leakage, the entire workflow is wrapped inside an isolated `ColumnTransformer` fitted strictly on training data ($N=54,909$, 80% split) and applied without modification to testing ($N=13,728$, 20% split) and production inference:

```
numeric_transformer = Pipeline([ ("imputer", SimpleImputer(strategy="median")), ("scaler", StandardScaler()), ]) categorical_transformer = Pipeline([ ("imputer", SimpleImputer(strategy="most_frequent")), ("encoder", OneHotEncoder(drop="first", sparse_output=False, handle_unknown="ignore")), ]) preprocessor = ColumnTransformer( transformers=[ ("num", numeric_transformer, ["age", "height", "weight", "ap_hi", "ap_lo", "bmi", "pulse_pressure", "map", "smoke", "alco", "active"]), ("cat", categorical_transformer, ["gender", "cholesterol", "gluc"]), ] )
```

4. Model Benchmarking, Composite Scoring & Calibration

Six distinct classification paradigms were trained and evaluated using **5-Fold Stratified Cross-Validation** on the training partition followed by held-out test set evaluation:

ALGORITHM	TEST ACC	PRECISION	RECALL (SENS)	F1-SCORE	TEST ROC-AUC	5-FOLD CV ACC	CV STD (±)	COMPOSITE SCORE
HistGradientBoosting 🏆	**0.7332**	**0.7538**	**0.6840**	**0.7172**	**0.8045**	**0.7345**	**±0.0010**	0.7880 (Selected)
Random Forest	0.7297	0.7494	0.6813	0.7138	0.8008	0.7308	±0.0029	0.7816
Logistic Regression	0.7262	0.7519	0.6663	0.7065	0.7960	0.7287	±0.0035	0.7730
Decision Tree	0.7268	0.7480	0.6752	0.7097	0.7836	0.7212	±0.0042	0.7712
K-Nearest Neighbors	0.7076	0.7116	0.6875	0.6994	0.7594	0.7052	±0.0032	0.7675
Support Vector Machine (RBF)	0.5479	0.5520	0.4568	0.4999	0.5709	0.4338	±0.0352	0.5379

4.1 Multi-Objective Selection Formula

The champion model was selected via a weighted clinical utility composite formula penalizing cross-validation variance: $$ ext{Composite} = (ext{Recall} imes 0.3) + (ext{ROC-AUC} imes 0.3) + ((1.0 - \min(10 imes \sigma_{ ext{CV}}, 1.0)) imes 0.2) + (ext{F1} imes 0.2)$$

4.2 Isotonic Probability Calibration

To ensure raw classification scores reflect true posterior disease probability, the champion model underwent 5-fold cross-validated **isotonic calibration** (`CalibratedClassifierCV(method='isotonic')`). Calibration refined test ROC-AUC to **0.8053**, producing calibrated probabilities where an output of 0.85 matches an empirical 85% positive rate.

5. Explainability Architecture: Global Permutation & Local Attributions

5.1 Global Permutation Importance

Computed across 10 shuffle iterations on held-out test data measuring the mean decrease in ROC-AUC when each feature is corrupted:

RANK	FEATURE	ROC-AUC DROP
1	Systolic BP (`ap_hi`)	**+0.1561**
2	Age	**+0.0391**
3	Cholesterol Level	**+0.0291**
4	Mean Arterial Pressure (`map`)	**+0.0044**
5	Body Mass Index (`bmi`)	**+0.0020**
6	Physical Activity (`active`)	**+0.0019**
7	Body Weight	**+0.0017**
8	Glucose Level	**+0.0015**

5.2 Real-Time Local Attribution

For individual patient inference, the backend executes **baseline perturbation attribution** against cohort medians:

Mathematical Formulation: $$\Delta p_i = P(Y=1 \mid \mathbf{x}) - P(Y=1 \mid \mathbf{x}_{i \leftarrow \text{median}_i})$$

- **$\Delta p_i > +0.001$:** Risk-Increasing Factor (red highlight in UI).
- **$\Delta p_i < -0.001$:** Protective Factor (emerald highlight in UI).
- **$|\Delta p_i| \le 0.001$:** Neutral / Baseline Factor.

Crucial Disclosure: Feature attributions represent mathematical model weight shifts, NOT physical anatomical lesion depth.

6. 3D Anatomical Engine & Encyclopaedia Britannica Diagram Architecture

The visual centerpiece of CardioVision AI is an interactive 3D WebGL anatomical human heart viewer. Following clinical feedback, the viewer eliminates oversized floating pins and strictly implements the **Encyclopaedia Britannica Anatomical Diagram** layout:

Left Margin Column (Encyclopaedia Britannica)

- **superior vena cava** — *(carries blood to heart)*
- **right auricle** — *(muscular pouch of right atrium)*
- **right atrium** — *(receives deoxygenated blood)*
- **right coronary artery (RCA)** — *(perfuses RV & nodal tissue)*
- **right ventricle** — *(pumps blood to lungs)*
- **inferior vena cava** — *(carries blood to heart)*

Right Margin Column (Encyclopaedia Britannica)

- **aorta** — *(carries blood to body)*
- **pulmonary trunk** — *(carries blood to lungs)*
- **left auricle** — *(muscular pouch of left atrium)*
- **pulmonary veins** — *(carry blood from lungs)*
- **LAD coronary artery** — *(perfuses anterior LV & septum)*
- **left atrium** — *(receives oxygenated blood)*
- **left ventricle** — *(pumps oxygenated blood to body)*

6.1 Coordinate Projection & SVG Leader Lines Engine

Rather than rendering HTML components inside 3D space (which blow up with zoom and occlude the organ), CardioVision AI runs a per-frame mathematical projection in `useFrame`:

```
// Projects 3D world vertex to 2D screen viewport coordinates: const worldPos = new
THREE.Vector3(...landmark.surfacePoint).applyMatrix4(heartMesh.matrixWorld); worldPos.project(camera); const
screenX = ((worldPos.x + 1) / 2) * canvasWidth; const screenY = ((-worldPos.y + 1) / 2) * canvasHeight; //
Normal vector dot product calculates occlusion when heart rotates to posterior: const isFacingFront =
normal.dot(camDirection) > -0.25;
```

Clean angled SVG leader lines connect margin labels horizontally to `(screenX, screenY)` on the heart surface, terminating in an anchor dot (•). When the heart rotates 360°, leader lines track surface vertices smoothly. Lines facing the posterior side gracefully dim.

6.2 Active 360° Turntable Rotation & View Presets

The heart model rotates continuously on its vertical axis ($\Delta heta = \Delta t imes 0.45 ext{ rad/s}$), providing complete 360° visibility of anterior ventricles, lateral sulci, and posterior pulmonary venous inflows. Users can pause/resume rotation or click 1-click angle presets: **Front (Anterior)**, **Left Side (LCx)**, **Back (Posterior)**, and **Right Side (RCA)**.

7. "Where is the Problem for the Patient?" — Clinical Vulnerability Mapping

To answer the clinical question: *"Where exactly does this specific patient have a problem?"*, CardioVision AI implements a deterministic physiological rules engine that translates patient vitals into concrete anatomical stress hotspots:

VITALS THRESHOLD	ANATOMICAL TARGET	PATHOLOGY TITLE	UNDERLYING HEMODYNAMIC MECHANISM
Systolic BP ≥ 140 mmHg	Left Ventricle & Ascending Aorta	High Afterload & Wall Tension	Left ventricle must contract against high peripheral vascular resistance, inducing chronic left ventricular hypertrophy (LVH) and shear fatigue along the aortic root.
Cholesterol Level 2 or 3	LAD & RCA Coronary Arteries	Atheroma Plaque Vulnerability	Excess circulating lipids accelerate subendothelial LDL accumulation and atheroma growth at high-shear bifurcations, particularly the Left Anterior Descending branch.
Pulse Pressure ≥ 60 mmHg	Aortic Arch	Aortic Elastic Stiffening	Loss of aortic elastance impairs the Windkessel buffering effect, exposing downstream coronary capillaries to unbuffered pulsatile pressure spikes.
Active Smoking (smoke=1)	Coronary Microvasculature	Endothelial Vasospasm	Combustion oxidants degrade endothelial nitric oxide bioavailability, provoking microvascular spasm and hypercoagulability.

Interactive Focus in UI: Clicking any identified problem badge immediately stops turntable rotation, snaps the 3D heart directly toward the affected anatomical structure, activates a pulsing red `STRESS ZONE` ring, and displays the underlying pathophysiological breakdown card.

8. Production FastAPI Backend Specifications

The backend exposes seven strictly validated RESTful endpoints operating at sub-15ms inference latency:

METHOD & PATH	INPUT SCHEMA	OUTPUT SCHEMA	DESCRIPTION & CLINICAL LOGIC
`GET /health`	None	`{status, model_loaded, version}`	Liveness probe verifying model persistence in memory.
`GET /dataset-summary`	None	`DatasetSummaryResponse`	Returns actual sample size (68,637), feature bounds, and balance.
`GET /model-performance`	None	`MetricsResponse`	Returns 6-model cross-validation metrics, confusion matrix, ROC data.
`GET /demo-patients`	None	`List[DemoPatient]`	Returns 3 empirical test-set profiles (Low 16%, Mod 46%, High 85%).
`GET /input-schema`	None	`InputSchemaResponse`	Form field specifications, units, physiological ranges, and labels.
`POST /predict`	`PatientInput` (Pydantic)	`PredictionResult`	Executes calibrated inference; returns risk index (0–100) and category.
`POST /explain`	`PatientInput` (Pydantic)	`ExplanationResult`	Performs perturbation analysis returning risk factors and attributions.

9. Clinical Governance & Mandatory Medical Disclaimers

Regulatory Classification: Research & Educational Prototype Only

CardioVision AI is engineered strictly as an educational and scientific communication prototype developed for hackathon demonstration. It has not undergone clinical trials, FDA 510(k) premarket notification, or CE mark clearance under EU MDR 2017/745 for Software as a Medical Device (SaMD).

- **Non-Diagnostic Status:** Predictions and 3D visual mappings generated by this software must NEVER be used to formulate medical diagnoses, establish treatment courses, or triage emergency care.
- **Human-in-the-Loop Imperative:** In any potential clinical decision support (CDS) deployment, licensed cardiologists retain complete diagnostic autonomy. The software functions solely as an adjunct educational communication tool.
- **Demographic Boundaries:** The training data originates from an Eastern European cohort aged 29–65. Generalization across pediatric, geriatric (>65), and ethnically diverse demographics requires prospective external multi-center validation.

MANDATORY CLINICAL DISCLAIMER

THIS SYSTEM IS AN AI-BASED RESEARCH AND EDUCATIONAL PROTOTYPE AND IS NOT A MEDICAL DIAGNOSTIC TOOL. PREDICTIONS AND 3D VISUALIZATIONS GENERATED BY CARDIOVISION AI SHOULD NEVER BE USED TO MAKE CLINICAL DECISIONS. CONSULT A QUALIFIED HEALTHCARE PROFESSIONAL FOR CLINICAL EVALUATION AND CARE.